

Engineering Sprint Plan — Week 8 (17 February 2026)

Sprint Overview

Sprint Duration: 10–21 February 2026 (Week 8)
Sprint Goal: Deploy Billing Validation Rule improvements to reduce merchant chargeback risk
Team: Platform Engineering (5 engineers)
Status: Deployed (Completed)

Key Changes

Change Request: CR-2987
Title: Billing Validation Rule (BVR) Enhancement
Priority: HIGH
Ticket Owner: Platform Team Lead

Description:
Implement stricter validation checks on API_Direct payment processing to reduce false positives and chargeback risk. The new rule requires:

- Merchant Account Status:** Active and not blacklisted (Rule A)
 - Transaction Amount Limits:** Within merchant's daily/monthly caps (Rule B)
 - Rate Limiting / Fraud Detection:** Transaction frequency and pattern analysis (Rule C)
 - Billing Validation Rule (Rule D):** NEW — Stricter merchant reserve requirements
 - Requirement: `account_balance > transaction_amount × 1.5` (150% reserve requirement)
 - Billing address: Must be on file (NOT NULL)
 - Failed transaction history: No more than 5 failed transactions in prior 7 days
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Deployment Timeline

Date	Time	Action	Status
17 Feb	10:30 AM	Code review + approval	✅ Complete
17 Feb	2:00 PM	Staging deployment (10% traffic)	✅ Complete
17 Feb	3:30 PM	Monitoring + alerts (30 min)	✅ Complete
18 Feb	9:00 AM	Production rollout (50% traffic)	✅ Complete

Date	Time	Action	Status
18 Feb	2:00 PM	Full production (100% traffic)	✅ Complete

Expected Outcomes

Success Metrics:

- Reduce merchant chargeback rate by 8-12% (baseline: 2.1%)
- Maintain transaction success rate >95% for compliant merchants
- No API latency increase >50ms

Risk Mitigation:

- Staged rollout (10% → 50% → 100%)
- Automated rollback if chargeback rate increases >15%
- 24/7 monitoring during first 48 hours post-deploy

Actual Results (Post-Deployment)

⚠️ ISSUE DETECTED — Week 9 (18 February onwards)

Observation: Transaction failures spiked from 5.8% (baseline) to 12.1% starting 18 February.

Root Cause Analysis:

- BVR (Rule D) stricter-than-intended: 150% reserve requirement too aggressive for SME merchants
- API_Direct payment method disproportionately affected
- SME merchant segment failed at 37% rate (vs. 5.2% baseline)

Impact:

- 3,300+ additional failed transactions per day
- £49,000 weekly revenue impact (lost merchant confidence)
- 8-10 key merchant partners at churn risk
- Regulatory escalation: FCA transaction accuracy filing required by 15 April

Status: Under Investigation (Requires Remediation Decision by 4 April)

Questions for Engineering Team

1. **Was the 150% reserve threshold correctly implemented?**

- Did testing use SME merchant account balances?
- Should threshold be adjusted (e.g., 120% instead)?

2. **Should Rule D be reverted immediately?**

- Cost: £0
- Timeline: 1 hour
- Risk: Chargebacks may increase temporarily

3. **Can we adjust thresholds in-place?**

- Cost: £12,000 (engineering time)
- Timeline: 8 hours
- Risk: Moderate (some SMEs still fail)

4. **Should we add a manual override queue?**

- Cost: £45,000 (3 days)
- Timeline: 3 days
- Risk: Low (manual review catches edge cases)

Contact Information

Sprint Lead: [Name] — Platform Engineering

Change Manager: Rachel Osei (Delivery Lead, Kestrel Digital)

Escalation: Chief Product Officer (Marcus Lee)

Compliance: Head of Compliance (Priya Patel) — FCA filing required

Status: Deployed but under remediation review

Last Updated: Week 9, 25 March 2026

Escalation Level: HIGH — Merchant churn risk